Week 13 Practice

1) Basic commands for storage partitions

There are many commands available in Linux to deal with disk partitions like:

a. fdisk
b. sfdisk
c. cfdisk
d. lsblk
e. parted etc.,

1) lsblk

To view the available hard disks in our system, use the command **lsblk** or using **cat /proc/partitions.** Both commands will display the same results, but in different ways.

```
root@ubuntu:~# lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0
       7:0    0     4K  1 loop /snap/bare
loop1
       7:1    0  63.5M  1 loop /snap/core
loop2
       7:2    0  63.3M  1 loop /snap/core
loop3
       7:3    0  73.1M  1 loop /snap/core
loop4
       7:4    0 349.7M  1 loop /snap/gnom
loop5
       7:5    0 349.7M  1 loop /snap/gnom
loop6
       7:6    0 460.6M  1 loop /snap/gnom
loop7
       7:7    0  91.7M  1 loop /snap/gtk-
loop8
       7:8    0  65.2M  1 loop /snap/gtk-
loop9
       7:9    0  12.3M  1 loop /snap/snap
loop10
       7:10   0  53.2M  1 loop /snap/snap
loop11
       7:11   0    46M  1 loop /snap/snap
```

```
loop12
       7:12   0  53.2M  1 loop /snap/snap
sda    8:0    0    20G  0 disk
├─sda1
       8:1    0   512M  0 part /boot/efi
├─sda2
       8:2    0     1K  0 part
└─sda5
       8:5    0  19.5G  0 part /
sr0   11:0    1  1024M  0 rom
```

- From the above output, the system has one disk **sda.**
- **sda** is again partitioned as sda1, sda2 and sda5.

2) fdisk -l

The above command gives the location of the partition and other details i.e., /dev/sda etc.,

```
root@ubuntu:~# fdisk -l
Disk /dev/loop0: 4 KiB, 4096 bytes, 8 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/loop1: 63.46 MiB, 66535424 bytes, 129952 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sda: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x5077e17e

Device     Boot   Start      End  Sectors  Size Id Type
/dev/sda1  *       2048  1050623  1048576  512M  b W95 FAT32
/dev/sda2       1052670 41940991 40888322 19.5G  5 Extended
/dev/sda5       1052672 41940991 40888320 19.5G 83 Linux


Disk /dev/loop8: 65.22 MiB, 68378624 bytes, 133552 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

3) fdisk /dev/sda

- It will take us to a different console where we can use the partitioning specific commands.
- This command supports following flags;

  m -> help
  p -> print partition table
  n -> create new partition
  d -> delete partition
  q -> quit without writing
  w -> write to disk

```
root@ubuntu:~# fdisk /dev/sda

Welcome to fdisk (util-linux 2.34).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): m

Help:

  DOS (MBR)
   a   toggle a bootable flag
   b   edit nested BSD disklabel
   c   toggle the dos compatibility flag

  Generic
   d   delete a partition
   F   list free unpartitioned space
   l   list known partition types
   n   add a new partition
   p   print the partition table
   t   change a partition type
   v   verify the partition table
   i   print information about a partition
```

2) Install and configure LVM

- Logical Volume Manager (LVM) is used on Linux to manage hard drives and other storage devices.
- To install LVM on Ubuntu, Debian, and Linux Mint, use the following command,

sudo apt install lvm2 or **sudo dnf install lvm2**

Steps to Install LVM on Ubuntu

Step 1: Prerequisites

1. Running Ubuntu 20.04 LTS Server.
2. sudo or root access to run privileged command.
3. apt or apt-get utility available in the System.

Step 2: Update the Server

a. Before installing lvm2 package, first sync your package cache with the enabled source repository using **apt update** command.

b. If any package needs to be upgraded, then upgrade it by using **apt upgrade** command.

```
root@ubuntu:~# apt update
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [59.7 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [95.5 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [940 B]
Get:7 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [108 kB]
Get:8 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 DEP-11 Metadata [275 kB]
Get:9 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 DEP-11 Metadata [410 kB]
Get:10 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:11 http://us.archive.ubuntu.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [7,972 B]
Get:12 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [30.5 kB]
Fetched 1,216 kB in 4s (292 kB/s)

Reading package lists... Done
Building dependency tree
Reading state information... Done
126 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ubuntu:~#
```

Step 3: Install LVM Utilities

Next step is to install LVM utilities using **apt install lvm2** command as shown below. This will install lvm2 package along with all its required dependencies.

```
root@ubuntu:~# apt install lvm2
Reading package lists... Done
Building dependency tree
Reading state information... Done
lvm2 is already the newest version (2.03.07-1ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 126 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Setting up lvm2 (2.03.07-1ubuntu1) ...
update-initramfs: deferring update (trigger activated)
Failed to restart lvm2-lvmpolld.service: Unit lvm2-lvmpolld.socket is masked.
invoke-rc.d: initscript lvm2-lvmpolld, action "restart" failed.
● lvm2-lvmpolld.service - LVM2 poll daemon
     Loaded: loaded (/lib/systemd/system/lvm2-lvmpolld.service; static; vendor preset: enabled)
     Active: inactive (dead)
       Docs: man:lvmpolld(8)
dpkg: error processing package lvm2 (--configure):
 installed lvm2 package post-installation script subprocess returned error exit status 1
Processing triggers for initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: Generating /boot/initrd.img-5.15.0-72-generic
Errors were encountered while processing:
 lvm2
E: Sub-process /usr/bin/dpkg returned an error code (1)
```

Step 4: Verify Package Installation

You can verify the installation path and list of utilities available through lvm2 package using **dpkg -L lvm2** command as shown below

```
root@ubuntu:~# dpkg -L lvm2
/.
/etc
/etc/init.d
/etc/init.d/lvm2
/etc/init.d/lvm2-lvmpolld
/etc/lvm
/etc/lvm/lvm.conf
/etc/lvm/lvmlocal.conf
/etc/lvm/profile
/etc/lvm/profile/cache-mq.profile
/etc/lvm/profile/cache-smq.profile
/etc/lvm/profile/command_profile_template.profile
/etc/lvm/profile/lvmdbusd.profile
/etc/lvm/profile/metadata_profile_template.profile
/etc/lvm/profile/thin-generic.profile
/etc/lvm/profile/thin-performance.profile
/etc/lvm/profile/vdo-small.profile
/lib
/lib/systemd
/lib/systemd/system
/lib/systemd/system/blk-availability.service
/lib/systemd/system/lvm2-lvmpolld.service
/lib/systemd/system/lvm2-lvmpolld.socket
/lib/systemd/system/lvm2-monitor.service
/lib/systemd/system/lvm2-pvscan@.service
/lib/systemd/system-generators
```

Step 5: Check Utilities Version

After successful installation, check some of the utilities version by using below command.

```
root@ubuntu:~# pvcreate --version
  LVM version:     2.03.07(2) (2019-11-30)
  Library version: 1.02.167 (2019-11-30)
  Driver version:  4.45.0
```

```
root@ubuntu:~# vgcreate --version
  LVM version:     2.03.07(2) (2019-11-30)
  Library version: 1.02.167 (2019-11-30)
  Driver version:  4.45.0
```

```
root@ubuntu:~# lvcreate --version
  LVM version:     2.03.07(2) (2019-11-30)
  Library version: 1.02.167 (2019-11-30)
  Driver version:  4.45.0
```

Step 6: Uninstall LVM Utilities

In case you are not planning to use LVM any further or does not require LVM utilities anymore, then you can remove them by using **apt remove lvm2** command as shown below.

root@ubuntu:~# apt remove lvm2

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
dmeventd libaio1 libdevmapper-event1.02.1 liblvm2cmd2.03 libreadline5 thin-provisioning-tools
Use 'apt autoremove' to remove them.
The following packages will be REMOVED:
lvm2
0 upgraded, 0 newly installed, 1 to remove and 341 not upgraded.
After this operation, 3,953 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 183880 files and directories currently installed.)
Removing lvm2 (2.03.07-1ubuntu1) ...
Warning: Stopping lvm2-lvmpolld.service, but it can still be activated by:
lvm2-lvmpolld.socket
Processing triggers for man-db (2.9.1-1) ...
```